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REVIEW ARTICLE

Gut Microbiota in Health and Disease

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Abstract

The human intestinal microbiota refers to the total population of micro-organisms that colonise the human intestines. Most organisms in the gut are non-pathogenic and co-inhabit with the host enterocytes in a commensalic and mutualistic relationship. The gut commensals aid in nutrient metabolism, xenobiotic and drug metabolism, antimicrobial protection, immunomodulation and maintenance of the integrity of the gut barrier and structure of the gastro-intestinal tract. Dysbiosis of these organisms have been implicated in the pathogenesis of several diseases. The advent of culture independent techniques has enabled the identification and phylogenetic classification of these organisms. These study methods include gene sequencing by high-throughput amplicon pyrosequencing (16sRNA for bacteria, accuracy 99.9%). Microbiota composition is affected by the mode of delivery, antibiotics usage during pregnancy, preterm labour, diet in infancy, environment & lifestyle, geographical locations, individual host genetics, in adults ageing & diet.

The dominant phyla (in a state of normal health) are Firmiculates (51%), Bacetroidetes (41%), Proteo bacteria (5%), Actinobacteria (1.8%) and Verrucomicrobia (1.2%). Most abundant fungi are Ascomycota and Microsporidia.The degree of archaeal diversity in the gut microbiota of healthy individuals appears to be low. The two genera Methanobrevibacter and Nitrososphaera appear to be mutually exclusive. Bacteriophages are the most predominant organism in the human gut microbiome. Role of gut microbiome is implicated in several diseases, which range from inflammatory bowel diseases to cancer, all the way to alzheimers. The understanding and recognition of various gut-microbiome axes has enabled elucidation of novel pathologic pathway that are ripe for therapeutic manipulation. Prebiotics and probiotics aid in the recolonization of the gut with normal "health-promoting-good" bacteria. Other therapeutic approaches are faecal microbial transplantation, which is being employed in clostridium difficile infection, and inflammatory bowel diseases. Their role and benefits in other diseases are emerging every day.

Introduction

The human intestinal microbiota refers to the entire population of micro-organisms that colonises the human intestines. It includes not just the bacteria, but also other organisms, such as fungi, archaea, viruses and protozoa. In recent years significant interest has evolved in the scientific community regarding the role of the normal gut flora in both health and diseases.

From an immunological perspective, most micro-organisms are considered potential or overtly pathogenic. However, most organisms in the gut are non-pathogenic and co-inhabit with the host enterocytes in a commensalic and mutualistic relationship.

The gut commensals aid in nutrient metabolism, xenobiotic and drug metabolism, antimicrobial protection, immunomodulation and maintenance

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Received : 24.07.2018
Revised : 26.11.2018
Accepted : 01.12.2018

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of the integrity of the gut barrier and structure of the gastro-intestinal tract[1].

Dysbiosis of these organisms have been implicated in the pathogenesis of several diseases. As such, regulation of this imbalance, is a novel treatment target presently being considered in many diseases.

Development of Gut Microbiota

Henry Tissier first postulated in 1899, that the placental barrier keeps foetuses sterile throughout a healthy pregnancy. Consequently, the presence of any bacteria in the uterus is generally considered as a potential danger for the foetus. However, recent studies indicate that the foetal gut is not sterile[2]. Pregnancy allows for increased physiological bacterial translocation, which selectively transports maternal microbiota to the mammary gland and to the foeto-placental interface. As such amniotic fluid and placenta harbour bacteria. When the foetus swallows the amniotic fluid the heirloom microbes derived from the maternal gut forms the first inoculum of intestinal microbiota.

Once born the human gut is rapidly colonised by an array of microbes derived during parturition from the maternal genital tract and also due to lactation[2,3]. The first colonisers are facultative anaerobes such as Streptococcus, Entero bacteriaceae and Staphylococcus. These organisms use the available oxygen and promote colonisation of strict anaerobes such as Bacteroides, Clostridium and Bifidobacterium[3].

Adult microbiotal compositionis attained by 3 to 5 years of age[4]. The composition is relatively stable throughout adulthood.

Elderly have increased facultative anaerobes like proteobacteria and bacilli. They also have decreased Faecalibacteriumprausnitzii and Clostridium cluster XIVa bacteria. These changes may be related to a decline in general health[5].

Centenarians had different and less diverse microbiota than adults and younger elderly people. They had decreased levels of Bacteroides, Bifido bacterium and Enterobacterieceae and increased levels of Clostridium. The therapeutic implications of these changes are still unknown[5].

Study Techniques

The advent of culture independent techniques have enabled the identification and phylogenetic classification of these organisms.

Gene sequencing : Molecular phylogenetics-In bacteria, the 16s rRNA subunits contains highly conserved regions of base sequences that reflect an absence of evolutionary changes[6]. Sequencing of 16sRNA, therefore, allows to identify and classify the organisms. Of the various methods commonly employed, high throughput amplicon pyrosequencing is the most commonly used[7]. It has an accuracy of 99.9%[1]. The representation of Archae in human microbial communities is less well defined than that of Bacteria, in part due to the difficulty in optimising the design of PCR primers that specifically target conserved regions of archaeal (versus bacterial) 16s rRNA genes[8].

Characterizing viral diversity requires different approaches : because no single gene is found in all viruses, no universal phylogenetic "barcode of life" equivalent to the SSU rRNA gene exists[9]. One approach has been to selectively purify virus-like particles from community bio specimens, amplify the small amounts of DNA that are recovered, and randomly fragment the DNA and sequence the fragments (shotgun sequencing).

Composition of the normal Gut Microbiota

The composition of the bacteria in human gut vary not only by the region of the gut[10], but also other factors like age, diet, lifestyle, geography, health state and diseases and antibioticsuse[10]. Therefore, there is a lot of flux in the composition at any given time for any individuals. However, large studies have

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shown that the dominant phyla (in a state of normal health) are Firmiculates (51%), Bacetroidetes (41%), Proteobacteria (5%), Actinobacteria (1.8%) and Verrucomicrobia (1.2%).

The dominant genera are : Faecalibacterium, Bacteroides, Clostridium, Eubacterium, ruminococcus, Peptococcus, peptostreptococcus, Bifidobacterium[11]. In healthy adults, Faecali bacteriumprausnitzii represent more than 5% of the bacteria in the intestine, making it one of the most common gut bacteria.

Due to the high acidity of the stomach most micro-organisms cannot survive. Less than 103 CFU/ml are seen. However, certain bacteria can thrive despite the adverse conditions. Helicobacter pylori is the commonest microbe present in the stomach[12] and determines the entire microbial landscape of the gastric flora, i.e, when Helicobacter pylori (H. pylori) inhabits the stomach as a commensal, there is a rich diversity constituted by other dominant genus of Streptococcus, Staphylococcus, Lactobacillus, Peptostreptococcus, Prevotella, veillonella and Rothia. This diversity rapidly shrinks once H. pylori acquires a pathogenic phenotype[12].

In the small intestine, trace amounts of micro organisms (103-105 CFU/ml) are found due to the proximity and influence of the stomach acids. Gram positive cocci and rod-shaped bacteria are the predominant micro-organisms. In the distal portions of the small intestine, the alkaline conditions support gram-positive bacteria like Entero bacteriaceae. Proximally facultative bacteria and distally obligate anaerobes predominate. The commonest organisms are Streptococci, Clostridium and Veillonella[11].

The large intestine contains the highest concentration of microorganisms. Ileum and caecum (103-109 CFU/ml) and colon (1010-1012 CFU/ml) provide a rich harbour for a wide variety of organisms. About 300 to 1000 species are found. 60% of the dry mass of faeces is composed of bacteria. 99% of the total composition comes from 30 to 40 species. 99% of the bacteria are obligate anaerobes[11,13]. Faecalibacteriumprausnitzii (a Firmiculate) is the most common species in healthy adults. Traditionally the Firmiculates: Bacteroidetes ratio has been implicated in predis position to disease states. The abundance of the phylum Proteobacteria is markedly low; and its absence along with high abundance of signature genera such as Bacteroides, Prevotella and Ruminococcus suggests a healthy gut microbiota[13]. The representation of fungi is significantly lower in individuals living in Westernized societies than in those living in non-Western societies. The most abundant fungal sequences belong to the phylum-level taxa Ascomycota and Microsporidia[8]. The phyla Ascomycota and Basidiomycota appear to be mutually exclusive. The most prevalent types of fungi are Saccharomyces (89%), Candida (57%) and Cladosporium (42%). The presence of Candida correlates with recent consumption of carbo hydrates and negatively total saturated fatty acids. Aspergillus is negatively correlated with short chain fatty acids. Saccharomyces does not appear to be affected by diet. Therefore, diet is the predominant influence of fungal composition.

The degree of archaeal diversity in the gut microbiota of healthy individuals appears to be low[14]. The two genera Methanobrevibacter and Nitrososphaera appear to be mutually exclusive. 25 to 95% of human guts are colonised by Methano brevibacter genera. Methanobrevibactersmithii is the most frequently occurring Archaea followed by Methanosphaerastadmanae[14]. Carbohydrate rich diet supports Methanobrevibacter.

Even though Bacteriophages are the most pre dominant organism in the human gut microbiome, the concept of the existence of a "gut virome" is paradoxically, very recent. The human gut harbours Bacteriophages (constituting 90% of the virome), microviruses (<145kb), small quantities of known pathogenic virus, plant-derived virus and giant virus (>300kb)[9].

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Bacteriophages were one of the first micro biological entities characterised in the literature. They can quite literally be defined as the "viruses of bacteria". They act as bacterial parasites[9], who inject their genome in their hosts, integrating with its genetic material (prophage state) and inducing other phage particle synthesis with bacterial cell lysis (lytic state).

Microviridae are small icosahedral viruses with circular single-stranded DNA genomes. Rotavirus, Astrovirus, Calcivirus, Norovirus, hepatitis E virus, Coronavirus, Torovirus, Adenoviridae, Picornaviridae and Reoviridae are common pathogenic organisms found in small quantities of the healthy gut[9].

In the past decades microbiological investigations have further discovered viruses infecting human intestinal parasites such as amebae (e.g. Mimiviridae, Mamaviridae, Marseilleviridae)[15] from cooling towers, rivers, lakes, and seawater. These giant viruses infect opportunistically pathogenic human parasites. In concerted effort with helper and satellite viruses and the parasitic host, they act as virophages to many known pathogenic viruses[9].

More recently plant viruses such as pepper mild mottle virus (PMMV), oat blue dwarf virus, grape vine asteroid mosaic- associated virus, maize chlorotic mottle virus, oat chlorotic stunt virus, panicum mosaic virus, and tobacco mosaic virus have been described.

Their biological importance depends on their concomitant presence in plants as pathogens and as "commensals" in human faeces, explaining how food intake has conditioned and continues to condition human gut virome composition and enterocyte life-cycle[9].

There are various non-pathogenic intestinal protozoa present in the human gut, like Chilomas tixmesnili, Edolimax nana, various species of Entamoeba and Iodamoebabuetschlii. These organisms do not harm the human body. Even in immunocompromised state. Symptomatic people who are found to have these protozoa in their stool should be examined for other causes of their symptoms[16].

Enterotypes

An enterotype is a classification of living organisms based on its bacteriological ecosystem in the gut microbiome[17]. Enterotypes are not dictated by age, gender, body weight, or national divisions. There are indications that long-term diet influences enterotypes[17]. So far three distinct enterotypes have been found. Each of these three enterotypes are identifiable by the variation in the levels of one of three genera: Bacteroides (enterotype 1), Prevotella (enterotype 2) and Ruminococcus (enterotype 3)[17].

Factors affecting Gut Microbiota Composition

The mode of delivery (normal vs caesarean), antibiotics usage during pregnancy, preterm labour, diet in infancy (breast feeding vs formula), environment and lifestyle, geographical locations, individual host genetics (mucus production rate, bile metabolism, host-derived antimicrobial peptides, hormones) are the initial factors affecting composition of the human microbiota during infancy[10]. Factors that influence during adults are ageing, as discussed earlier, diet, dietary poly phenols, residual pesticides and insecticides in food and consumption of animal products treated with antibiotics and other drugs[10].

Functional aspects of the normal Gut Microbiota

The gut microbiota maintains a symbiotic relation ship with the gut mucosa and imparts substantial metabolic, immunological, and gut protective functions in the healthy individual. The principal function of the gut microbiota are : i) Nutrient metabolism, ii) Xenobiotic and drug metabolism, iii) Antimicrobial protection, iv) Immunomodulation and v) Maintenance of the integrity of the gut barrier and structure of the GIT.

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Carbohydrate metabolism : The gut microbiota derive their nutrients from dietary carbohydrates. Saccharolytic fermentation of carbohydrates that escaped proximal digestion and indigestible oligosaccharides by colonic organisms result in the synthesis of short chain fatty acids (SCFAs). These SCFAs like acetate, butyrate and propionate are rich sources of energy for the host. Organisms like Bacteroides, Roseburia, Bifidobacterium, Faecali bacterium and Enterobacteria ferment carbo hydrates. Syntropism exists between fungi, bacteria and archae to enable carbohydrate metabolism[8].

Protein metabolism: gut microbes impact nitrogen balance by de novo synthesis of amino acids and intestinal urea recycling. Upto 20% of the circulating lysine, leucine and threonine are synthesised by gut microbiota with substrates derived from dietary carbohydrate metabolism. The gut microbiota also participate in urea nitrogen salvaging.

Lipid metabolism : The microbiota has a positive impact on lipid metabolism. Suppression of Fiaf (fasting-induced adipocyte factor) increases hepatic lipogenesis with increased LPL activity in adipocytes through ChREBP- (carbohydrate responsive element binding protein) and SREBP (sterol regulatory element binding protein). This increases the triglyceride storage in adipocytes. Bacteroides thetaiotaomicron also regulates the expression of a colipase that is required by pancreatic lipase for lipid digestion[18].

The oxalate that is synthesised in the intestine during carbohydrate metabolism is countered by organisms like oxalobacterformigenes[19], lactobacillus, and bifidobacterium. Thereby reducing the risk of formation of oxalate stones in the kidney.

The microbiota also has a role in neurotransmitter[1] production (L-histidine to histamine and Glutamate to GABA), synthesis of vitamin k and some components of vitamin B, synthesis of conjugated linoleic acid (known to be anti-diabetic, antiathero genic, antiobesogenic, hypolipidemic, conversion of primary bile acids to secondary bile acids[20], and breakdown of polyphenolic compounds[1].

Intestinal microbiota has a major role in conversion of prodrugs (sulfasalazine to amino salicylate, L-dopa to dopamine), reduction of bio availability of certain drugs (digoxin by Eggerthellalenta) or increasing the bioavailability of drugs (acetaminophen, irinotecan)[1].

The gut microbiota keeps the luminal pathogenic microbes away from epithelial contact. The microflora also activate dendritic cells that induces plasma cells to express secretary IgA. The secretary IgA coats the gut microbiota rendering them resistant to degradation by bacterial proteases. This restricts the translocation of the microbiota from the intestinal lumen in to the circulation and prevents systemic immune response[1].

Gut microbiota contribute to immunomodulation in tandem with both the innate and adaptive immune systems by shaping a normal GALT, mediating effect T cell response by developing IL-7, develops T regulatory cells[21].

The gut microbiota help in maintenance of the integrity of the gut barrier/ structure of the GIT. Bacteroides maintains desmosomes at the epithelial villus, cell wall peptidoglycans of the microbiota maintain tight junctions, lactobacillus prevents cytokine induced intestinal epithelial cell apoptosis, and Akkermansiamuciniphilia decreases metabolic endotoxaemia[21].

Role of Microbiota in Diseases

Peptic Ulcer

Conditions that predispose to decrease in the acidity of the gastric mucosa enable the otherwise harmless, commensalicH. pylori to take up a pathologic role[12]. Once the urea secreting H. pylori burrows in to the epithelium, it induces increase in

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acidophilic Proteobacteria, Spirochaetes and Acidobacteria. It also causes marked decline in Actinobacteria.

Inflammatory Bowel Diseases

In ulcerativecolitis there is excess Proteobacteria and Actinobacteria and in Crohn's disease Entero coccus faecium and Proteobacteria are in excess. There is also reasonable evidence that correcting gut flora imbalances by taking probiotics with Lactobacilli and Bifidobacteria can reduce visceral pain and gut inflammation in IBD[22].

Irritable Bowel Syndrome

Irritable bowel syndrome is a result of stress and chronic activation of the HPA axis[23]. Overall there is decreased species diversity, with low levels of Lactobacilli and Bifidobacteria, high levels of facultative anaerobes like E. coli and increased Firmicutes : Bacteroides.

Obesity and Metabolic Syndrome

The gut flora has also been implicated in obesity and metabolic syndrome due to the key role it plays in the energy harvesting from food. The Western pattern diet, with its lack of fibres and simple sugars, reduces bacteroides and proportionately increase firmicutes. There is a dramatic fall in overall diversity with a bloom of a single class of firmicutes: the mollicutes[24].

This change alters the metabolic potential of the gut[24]. There is enrichment for phosphotransferase systems that import and ferment simple sugars, enrichment of genes encoding beta-fructosidases and depletes motility genes. Consequently, there is increased capacity to import "Western-diet"-typical carbohydrates and increased capacity to metabolise sugars to SCFAs[24].

Microbiota can also influence eating behaviours based on the preferences of the microbiota, which can lead to the host consuming more food eventually resulting in obesity[25]. It has generally been observed that with higher gut microbiome diversity, the microbiota will spend energy and resources on competing with other microbiota and less on manipulating the host[25]. The opposite is seen with lower gut microbiome diversity, and these micro biotas may work together for to create host food cravings.

Conversely, just as gut flora can function in a feedback loop that can drive the development of obesity, there is evidence that restricting intake of calories (i.e. dieting) can drive changes to the composition of the gut flora.

Liver Diseases

In cirrhosis: bacterial translocation, in association with the presence of vascular shunts, is responsible for increasing circulating levels of lipopolysaccharide (LPS) in the portal blood[20,26]. Selective increase in certain micro-organisms. LPS and pro-inflammatory cytokines result in over production of NO, following the activation of iNOS, eNOS (endo thelial) and nNOS (neuronal). This over production causes vascular alteration. Decreased conversion of primary to secondary bile acids due to the endotoxaemia[26].

In viral hepatitis

high levels of endotoxin cause overproduction of the pro-inflammatory cytokines including Tumour Necrosis Factor (TNF)[26]. In alcoholic liver disease: intestinal bacteria of the large bowel participate in the metabolism of alcohol, resulting in the introduction of high concentrations of toxic acetaldehyde into the lumen[26]. Acetalde hyde, per se, increases intestinal perme ability (gut leakiness), indirectly changing the microbiota equilibrium, and increasing the LPS quantity that arrives at the liver.

Cardiovascular Diseases

Certain dietary nutrients possessing a trime thylamine (TMA) moiety, namely choline/ phosphat

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idylcholine[27] and L-carnitine, participate in the development of atherosclerotic heart disease[27]. A meta-organismal pathway was elucidated involving gut microbiota dependent formation of TMA and host hepatic flavin monooxygenase 3-dependent (FMO3-dependent) formation of TMA-N-oxide (TMAO), a metabolite shown to be both mechanistically linked to atherosclerosis and whose levels are strongly linked to cardiovascular disease (CVD) risks[27].

Celiac Disease

Patients with celiac disease (CD) are found to have reduced number of Bifido bacterium spp., Lacto bacillus B. longum and increased numbers of Bacteroides spp[28]. There is also increase in enter obacteria and staphylococci in the gut of untreated CD[28]. The dysbiosis disappeared only after long-term therapy with gluten-free diet[28]. Recent interventional studies have shown that there may be a role of probiotics, like B. infantis B. longum, and B. breve in the therapeutic arsenal of CD.

Rheumatoid Arthritis

Normal intestinal microbiota in patients with rheumatoid arthritis (RA) may harbour, for genetic reasons, bacteria with cell walls capable of inducing arthritis. In susceptible subjects, with continuous seeding of bacterial products from the gut, erosion, exposition of cartilage antigens, and self-perpetuating chronic arthritis follow the synovial inflammation. The RA patients had significantly less bifidobacteria and bacteria of the Bacteroides-Porphyromonas Prevotella group[29], Leptotricha, Bacteroides fragilis subgroup, and Eubacteriumrectale Clostridium coccoides group[29]. This dysbiosis leads to the interplay between pre disposing genetic factors & environmental triggers which is required for the manifestations of RA.

Bronchial Asthma & Atopy

A growing body of epidemiological and micro biological literature support the hypothesis that the genesis of allergic disease and by extension asthma development, may lie, at least in part, in the communities of microbes that exist in the gastrointestinal tract[30]. Early life antimicrobial exposure, Caesarian birth[1], formula feeding, lack of maternal exposure to pets or livestock during pregnancy, and maternal consumption of anti microbials during pregnancypre disposes a child to bronchial asthma.

Type 1 Diabetes Mellitus

Autoimmune T-cell mediated destruction of insulin -producing -cells is the pathological hallmark of type 1 diabetes. Murine models lacking My D88 protein (an adaptor for multiple innate immune receptors that recognise microbial stimuli) develop robust diabetes unless they are concurrently provided with a defined consortium of gut microbes found in normal human guts[21].

Type 2 Diabetes mellitus

type-2-diabetes patients were characterised by a moderate degree of gut microbial dysbiosis[31]. There is a decrease in the abundance of some universal butyrate-producing bacteria and an increase in various opportunistic pathogens, as well as an enrichment of other microbial functions conferring sulphate reduction and oxidative stress resistance[31].

Cancer

The role of H. pylori for gastric and duodenal cancer is perhaps the most prominent[12]. Increase in proteobacteria has been linked to colorectal cancer. As such Lactobacillus, streptococcus and Bifidobacterium are considered to be anti-colorectal -tumoregenic[18]. Increased methylobacterium radiotolerans and decreased sphingomona syanoikuyae has been studied in breast cancer models[32]. TLR4 and the intestinal microbiota are not required for Hepatocellular carcinoma (HCC) initiation but for HCC promotion, mediating increased proliferation, expression of the hepato mitogen epiregulin, and prevention of apoptosis[33].

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Gut sterilization restricted to late stages of hepato carcinogenesis reduced HCC, suggesting that the intestinal microbiota and TLR4 represent the rapeutic targets for HCC prevention in advanced liver disease[33].

Psychiatric Illnesses

Hypothalamic pituitary axis (HPA) hyperactivity characterises depression, and this hyperactivity may be mediated by a proinflammatory reaction to stress (high levels of IL-1, IL-6)[23]. Several selective serotonin reuptake inhibitors (SSRIs) have a negative immunoregulatory effects by increasing relative amounts of IL-10, a cytokine that suppresses inflammation[34]. Similar immunoregulatory effects of probiotic organisms are thought to be effected by the synthesis of IL-10 and thereby probiotics might have antidepressant effects[34].

A hypothesis proposed that stress and antibiotics are the mechanisms by which gut microbiota are altered before the onset of obsessive compulsive disorder (OCD) symptoms[35]. OCD symptoms occasionally develop as a part of the PANDAS (Pediatric autoimmune neuropsychiatric disorders associated with streptococcal infections). It is currently believed that these symptoms are explained by group A beta-hemolytic streptococcal infections. However, the symptoms are instead caused by antibiotics that are administered as treatment of the infection, but inadvertently alter microbiota[35]. Furthermore, stressful life events known to trigger OCD, such as pregnancy, alter gut microbiota prior to onset of OCD symptoms. Suggested treatment for OCD would be the re-introduction of beneficial bacteria modifying the gut microbiome, thereby ameliorating OCD symptoms.

Multiple Sclerosis

In multiple sclerosis there is a reduction in Faecali bacteriumprausnitzii, ruminococcus, lacto bacillaceae, clostridium and increased abundance of psuedomonas, Mycoplana, Haemophilus, Blautia, and Dorea genera[36]. This dysbiosis allows initiation of a relapsing-remitting autoimmune disease driven by myelin-CD4+T cells[37]. The recruitment and activation of autoantibody producing B cells from the endogenous immune repertoire depends on availability of the target autoantigen, myelin oligodendrocyte glycoprotein (MOG), and commensal microbiota. These events trigger organ-specific autoimmune disease and these processes may offer novel therapeutic targets[37].

Autism-Spectrum Disease (ASD)

Evidence points towards maternal diet and inflammatory events resulting in systemic products of inflammation including cytokines, IL-6 and IL-17A, that can alter central nervous system neurodevelopment of regions critical to the cognition of social interaction[38]. Select commensalic microbiota can induce or attenuate these molecules. Compared with ASD-unaffected children, children with ASD had increased mucosal tissue levels of select microbial taxa, mainly members of the genus Clostridium[38] along with significant increase in the Firmicutes/Bacteroidetes ratio in autistic subjects due to a reduction of the Bacteroidetes relative abundance. There was also a decrease in the relative abundance of Alistipes, Bilophila, Dialister, Parabacteroides, and Veillonella, while Collinsella, Corynebacterium, Dorea, and Lactobacillus were significantly increased.

Parkisnsonism

It is noted that for the manifestations of parkinsonism, both dysbiotic microbiota and disease-competent microglia are required in concert. Gut derived SCFAs modulate microglial actions. Microglia further promote synuclein mediated motor deficits and brain pathology[39].

Alzheimer's Disease

Insulin resistance in the brain has been recently associated with Alzheimer's disease (AD)[40]. There is growing evidence that suggests that the increased

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permeability of intestine and blood-brain-barrier induced by gut microbiota disturbances increases the incidence of neurodegeneration through inflammation. As such the onset of AD, supports the "hygiene hypothesis" theory and may in fact be intrinsically linked to the Westernisation of the human diet and its downstream effect of human gut microbiome. Therefore, Alzheimer's may be another manifestation of the metabolic disorder spectrum[40].

Prebiotics/Probiotics/Synbiotics

Prebiotics

Marcel roberfroid had first introduced the concept of prebiotics in 1995 which was later redefined in 2007. Prebiotics are selectively fermented food ingredient that allows specific changes, both in the composition and/or activity in the gastrointestinal microflora that confers benefits upon host well-being and health[22,41]. They conceptually inter mediate between food and drug. They potentiate the activity of bifidobacteria and lactobacillus.

Probiotics

Probiotics are defined as, "live microorganisms which, when administered in adequate amounts, confer a health benefit on the host."The introduction of the concept is generally attributed to Nobel recipient Elie Metchnikoff, who in 1907 postulated that yoghurt-consuming Bulgarian peasants lived longer lives because of this custom. Probiotics i) Secrete bacteriocins/defensins, ii) Competitive inhibition against pathogenic bacteria, iii) Inhibit bacterial adhesion/trans location, iii) Reduce luminal pH, and iv) Increase mucus secretion[22,26].

The fundamental concept of using these organisms in the treatment armamentarium is mimicking the physiological health promoting functions of the "good" bacteria. Probiotics are generally considered safe but caution is needed in patients with immuno deficiency, short bowel syndrome, central venous catheters, cardiac valve disease.

Prebiotic substances	Food sources rich in prebiotics	
o Trans-galacto-oligosaccharide	o Gum Arabic	o Leek
o Inulin	o Chicory	o Onion Asparagus
o Larch arabinogalactan (LAG)	o Jerusalem Artichoke	o Wheat bran
o Resistant starch	o Dandelion greens	o Whole wheat flour (cooked)
o Pectin	o Garlic	o Banana
o Beta-glucans		
o Xylo-oligosaccharides		

The recommended serving size is 4 to 8 grams/day.

Probiotic Organisms	Good Sources
o Lactobacillus casei	o Yoghurt
o Lactobacillus plantarum	o Pickle vegetables
o Lactobacillus bulgaris	o Fermented bean paste: tempeh, miso
o Lactobacillus acidophilus	o Kefir
o Bifidobacterium longum	o Buttermilk and karnemelk
o Bifidobacterium infantis	o Kimchi/ Páocái
o Streptococcus thermophiles	o Sauerkraut
o E. coli strain Nissle 1917	o Soy sauce and tamari

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Synbiotics

It is the combination of prebiotics and probiotics. The term is only applicable if the net health benefit is synergistic. Eg: Bifidobacteria and fructo-oligo saccharides, lactobacillus rhamnosus GG & inulins.

Therapeutic avenues and advances

Faecal Microbial Transplantation (FMT)

FMT involves the transplantation of gut microbiota from healthy individuals to a recipient with a disease. It restores the normal healthy colonic flora. The process can be achieved by endoscopy, colonoscopy, enema or to be taken as freeze-dried material in capsule form.

It has been found to be more effective than vancomycin alone for clostridium difficile infection (CDI). It has been experimentally used in colitis, constipation, IBS, multiple sclerosis and Parkinsonism. Although considered safe the possible side effects include sepsis, fever, exacerbation of IBD.

In 2012, a team of researchers from the Massachusetts Institute of Technology founded OpenBiome[42], the first public stool bank in the United States. OpenBiome provides clinicians with frozen, ready-to-administer stool samples for use in treating CDI, and supports clinical research into the use of faecal transfer for other indications.

CDI

Clostridium difficile infection causes dysbiosis. There is increase in proteobacteria and overall decrease in Firmicutes and Bacteroides. Apart from FMT, the other microbiota derived treatment strategies that are being studied are RBX2660(43), MET-1. SHP-01, SYN-004, AVIDOCINS, SER-109(44). These novel agents are in various phases of trial. Of special note is the drug Ridinilazole(45), which specifically targets CDI whilst sparing the normal gut microbiota.

Ulcerative Colitis/ Crohn's Disease

IBD causes decrease in firmicutes and increase in proteobacteria, bacteroidetes and bacilli. The investigational agents that are being studied are RBX-2660, SER-287, VE-202, CBM588[46], CNDO-201, Bacteroides thetaiotamicron, AG0-14, VT-301, SGM-1019.

Other Novel indications that are being explored are CBM588(46) for bacterial vaginosis and non alcoholic fatty liver disease, Lactin-V for recurrent urinary tract infections, Lactobacillus plantarum for cardiovascular diseases and obesity, Lactobacillus helviticus for rheumatoid arthritis, Blautix for infla matory bowel syndrome and NM504/505 for metformin-intolerant type-2-diabetes.

Conclusion

Increasing recognition of the profound role that intestinal microbiota play in our body in health and disease states is a novel and interesting therapeutic avenue. As more evidence and techniques emerge, the use and modulation of gut microbiota will become an important adjunct to a savvy physician's treatment armamentarium.

How this article to be cited : Shreyashi Ganguly, Chandrasekhara P . Gut Microbiota in Health and Disease. Journal of Internal Medicine. Research, Reviews, Reports 2019 ; 7(1) : 19-30

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FORM IV (See Rule 8)

1.	Place of Publication	The Association of Physicians of India Karnataka Chapter, Bengaluru 560052
2.	Periodicity	Quarterly
3.	Printers Name	Sri V Srikantha
	(Whether Citizen of India)	Indian Citizen
	Address	Chap Prints, 21/5, 8th Main, 3rd Cross Saraswathi puram, Mysuru 570009. Karnataka, India.
5.	Publishers Name	Dr M Premanath
	(Whether Citizen of India)	Indian Citizen
	Address	The Association of Physicians of India, Karnataka Chapter, API Bhavana,# 16/F, Millers Tank Bed Area, Vasanth Nagar Bengaluru 560052, Karnataka, India
6.	Editor's Name	Dr M Premanath
	(Whether Citizen of India)	Indian Citizen
	Address	Nisarga, #21, Block-9, Madhuvana layout Srirampura II stage, Mysuru, 570023. Karnataka, India
7.	Name and address of individuals Who own the news paper and Partners an share holders holding More than one percent of the capital	The association of Physicians of India Karnataka Chapter, Bengaluru 560052

I, Dr M PREMANATH, hereby declare that the particulars given above are true to the best of my knowledge and belief

25th December 2018

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Signature of the Publisher